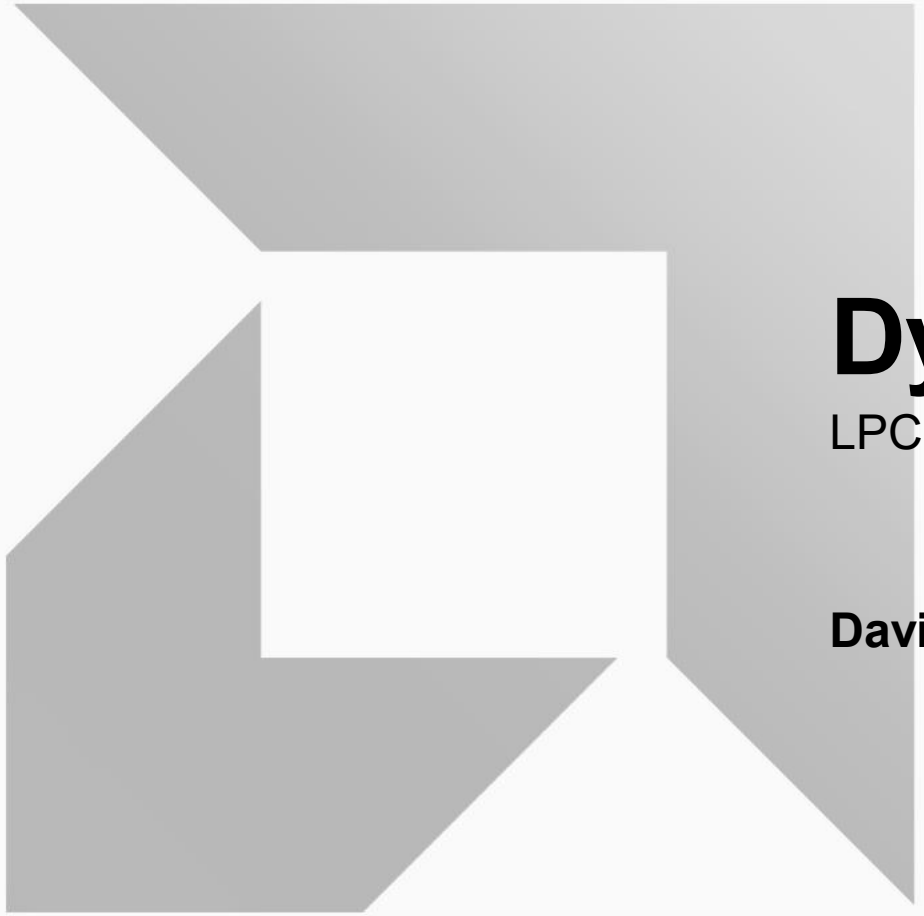


Dynamic Mitigations

LPC 2025 – x86 Micro-conference

David Kaplan, AMD

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Dynamic Mitigations

- Enables ***re-configuration of CPU mitigations at runtime*** through sysfs interface
  - e.g., `echo “mitigations=off” > /sys/devices/system/cpu/mitigations`

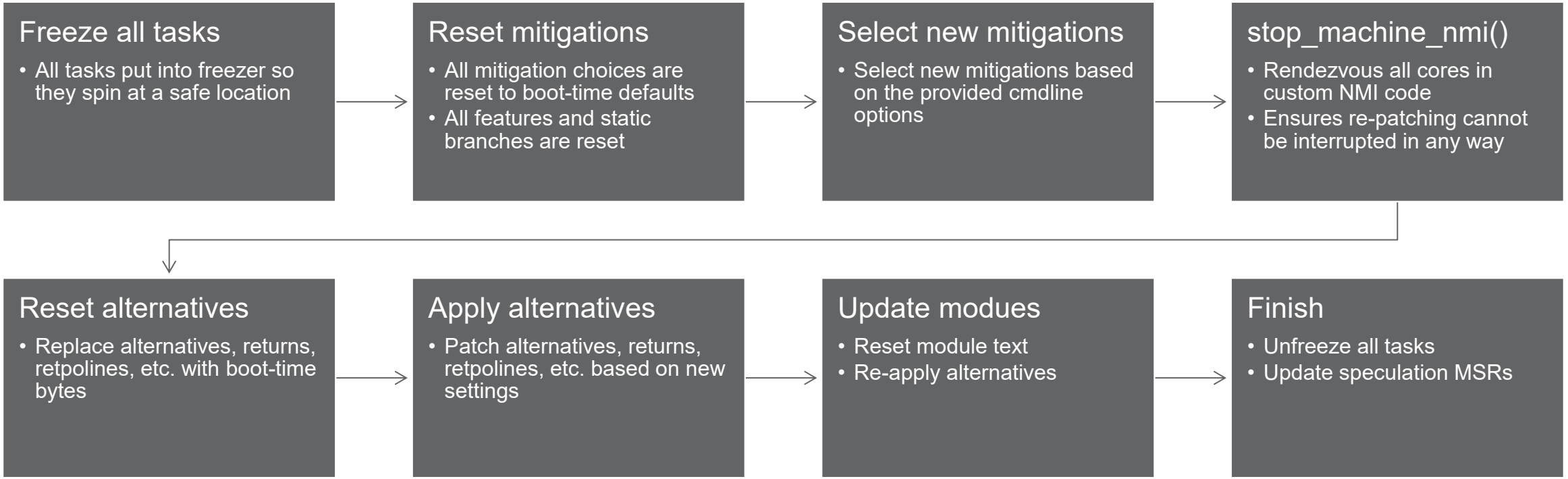
- **Runtime Policy**
  - Enable policy to be chosen later in boot after system determines its intended use
  - Some VMs start as generic shells, later customized after a network data file is fetched (e.g., cloud-init)
    - Today these always run with default settings (everything enabled), which is unneeded in many cases
  - Enables ability to turn off unneeded mitigations, recovering performance
- **Mitigation Changes**
  - Sometimes mitigation choices must be changed due to new security findings
  - Supports applying new choices without a reboot or kexec
- **Mitigation Testing**
  - More efficient testing of mitigations, especially when multiple options are available

- Re-patching takes ~50ms, expected to occur once as part of system initialization
- RFC link: https://lore.kernel.org/lkml/20251013143444.3999-1-david.kaplan@amd.com/

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How to Re-Patch



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